Deep learning: Portfolio assignment 2

Jeffrey Gerrits - Master of Applied Data Science UOS3 - 25-02-2026

This portfolio assignment focusses on the hypertuning of (convolutional) neural networks. After fiddling around with convolutions, pooling-, normalization- and dropout layers, a hypothesis was formed based on the expectation that a dropout later acts a regularization technique that reduces overfitting. The hypothesis is as follows:

H0: The addition of a dropout layer does not affect the generalization performance of the neural network.
H1: The addition of a dropout layer improves the generalization performance of the neural network.

To test this hypothesis, the training and test loss of different dropout settings were compared. The network consists of 3 convolutional layers to find patterns, where each layer is followed by ReLu and maxpool. The outputs go in to an AvgPool and Flatten layer. Then a dense network of two hidden layers, each followed by a ReLu and dropout layer, lead to the output layer of 10 classes. Three dropout settings were tested during the experiment: 0.0, 0.2 and 0.5.

In the first iteration the number of units of the hidden layers was 128 and 64. For a second iteration this was increased to 256 and 128 layers. Also the number of epochs was increased from 3 to 10.

Results

The first results (small network size) show an increase in training loss from 0.62 to 0.67 to 0.76 with the increasing dropout rate. The test loss also increases, but less significantly: 0.60 to 0.61 to 0.64. This means that the generalization performance did not improve. The results also indicate that there was little to no overfitting in the first place, so a dropout layer was not very useful for this configuration.

A larger network was used in the second iteration, but the outcomes still did not show signs of overfitting. With the dropout layer set at 0.0 the train- and test loss were almost equal. While the dropout rate increased, so did the train- and test loss. It is therefore concluded that, given the circumstances of this experiment, the performance of the network decreased after increasing the dropout rate. The addition of a dropout layer did not positively affect the generalization performance because training and test performance were already closely aligned. We therefore cannot reject the null hypothesis based on this experimental setup.

Figure 1 ML Flow line graphs for train- and test loss of the first experiment configuration

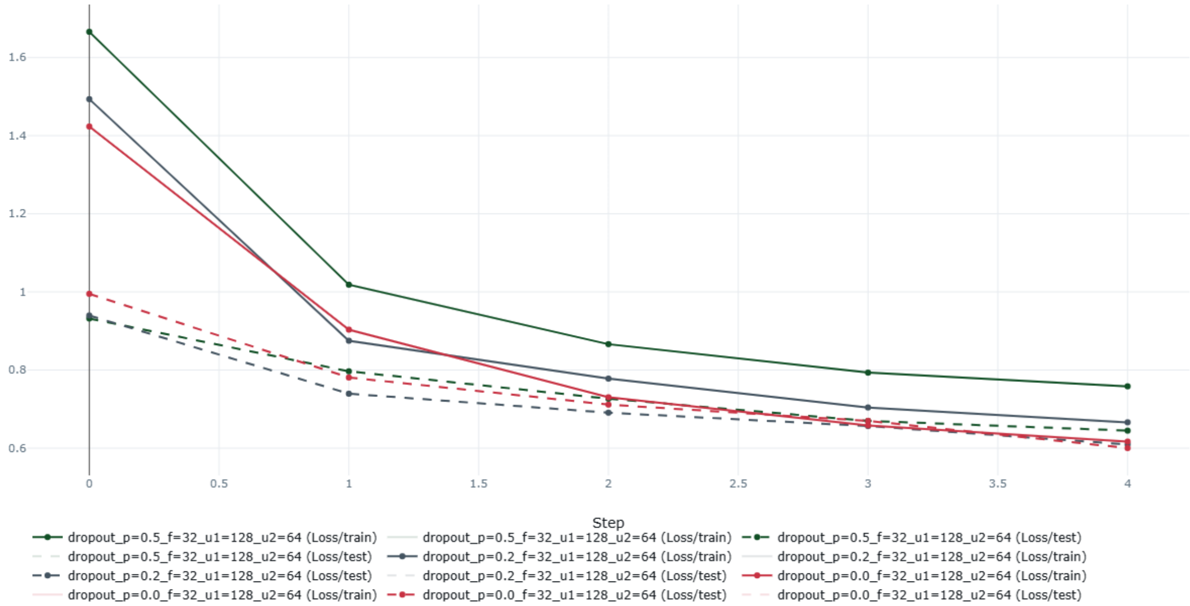

Figure 2 ML Flow line graphs for train- and test loss of the second experiment configuration